

NLP Automated Customer Reviews

From thousands of Amazon reviews to insights, summaries and German translation

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AI Engineering Bootcamp - Final Project

Live app: [Streamlit Community Cloud](#)

Project Overview: Data and Hypothesis

- **Dataset:** Amazon consumer product reviews (Datafiniti) - 34,625 reviews, 40 named products; product name, category, review text, star rating
- **Problem:** thousands of reviews across platforms cannot be read by hand
- **Hypothesis:** NLP can automatically (1) detect sentiment from text, (2) group products into categories and (3) summarise each category into buyer advice
- **Deliverables:** 5 tasks - sentiment, clustering, generative summaries, public dashboard, German translation

Structure of the Analysis

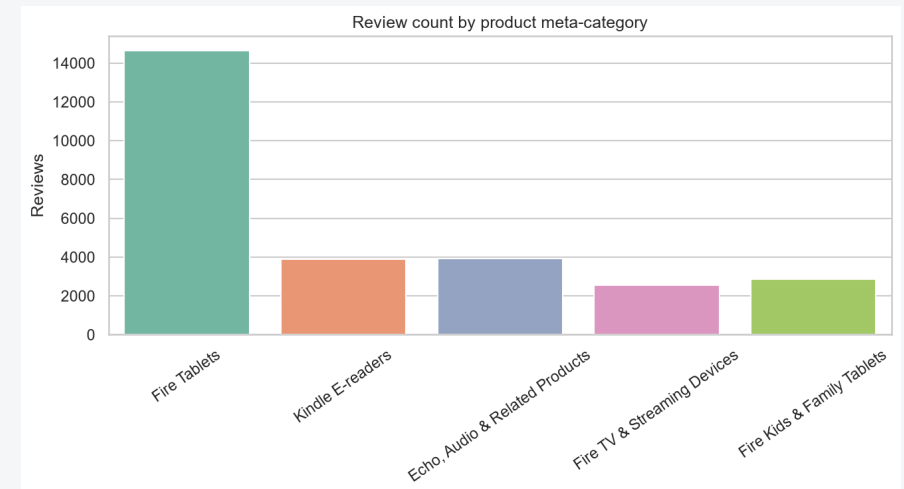
- **Clean** the data and label sentiment from stars (1-2 negative, 3 neutral, 4-5 positive)
- **Task 1** sentiment: TF-IDF + LinearSVC, chosen by macro F1
- **Task 2** clustering: K-Means, 5 product categories
- **Task 3** category summaries with DistilBART
- **Tasks 4-5** Streamlit app with dashboard, predictor and German translation

Data Wrangling and Cleaning

- **Challenges:** missing values, duplicate reviews, inconsistent product names, 92% positive ratings
- **Fixes:** dropped incomplete rows, normalised names, removed duplicates, cleaned the text
- **Enrichment:** no scraping or data APIs; we used the provided dataset plus pretrained Hugging Face models
- **Lesson:** stripped text hurt translation, so it uses the raw reviews

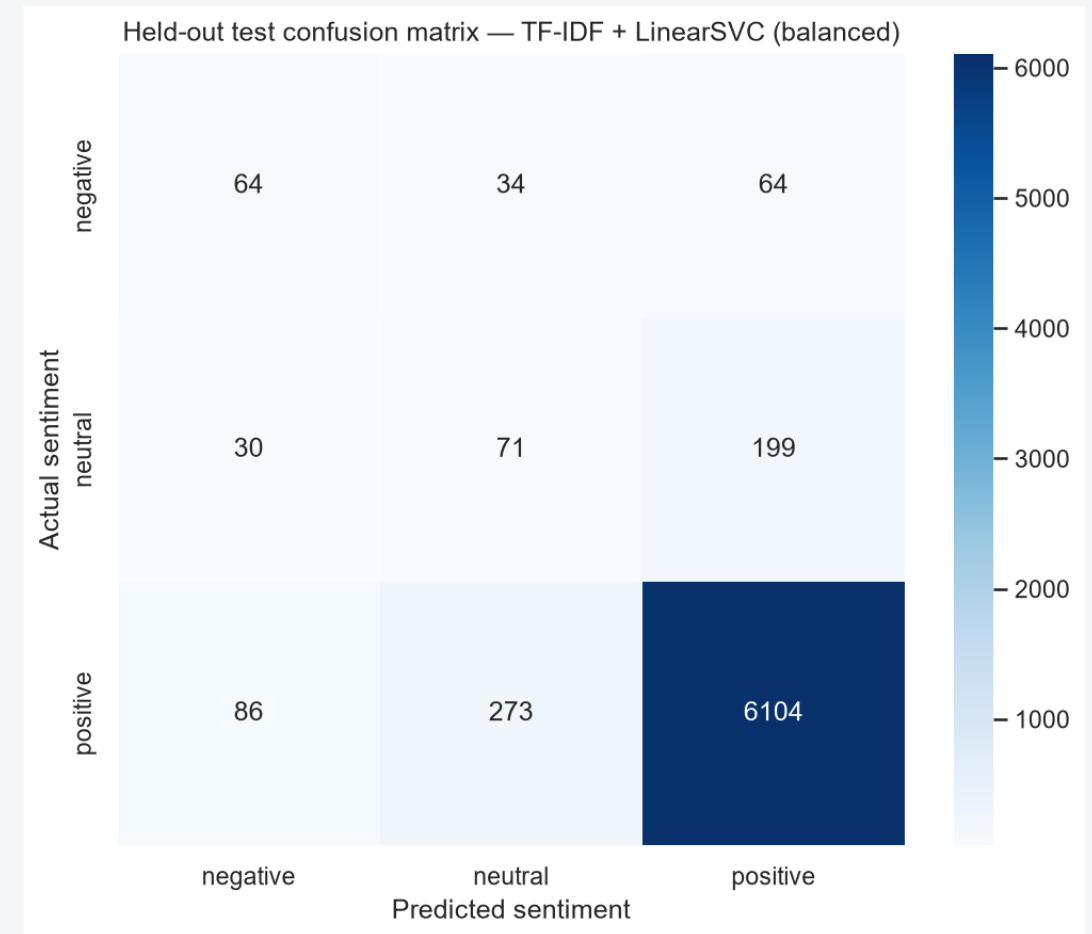
Exploratory Data Analysis

- **Methods:** label distribution, per-class metrics, confusion matrix, silhouette scores, cluster profiles, top terms
- **Imbalance:** about 92% of reviews are positive; average rating 4.58 / 5
- **Clusters:** Fire Tablets, Kindle E-readers, Echo and Audio, Fire TV, Fire Kids - silhouette ~ 0.11 (overlapping)
- **Complaint themes** from keyword matching on negative reviews, shown per category



Key Result: Sentiment Model

- Test set: 6,925 untouched reviews
- Accuracy **90.1%** - but macro F1 only **51.2%**
- F1: positive 95.2%, negative 37.4%, neutral 20.9%
- Many negative and neutral reviews are predicted as positive



Generative AI: Summaries and German Translation

- **Summaries:** DistilBART writes a buyer guide for each of the 5 categories
- **Translation:** English to German with MarianMT; product names stay unchanged; other languages are flagged
- **Evaluation (30 reviews):** round-trip BLEU ~55, ROUGE-1 ~0.83; sentiment kept in 80%
- **Example fixed:** 'tablet' was translated as the German word for pill

From Models to a Product: Live App Demo (video)

- **"We did not stop at building NLP models. We turned them into a public application that lets users analyse reviews, predict sentiment and translate customer feedback into German."**
- Three tabs: insights dashboard, single-review predictor, Translate to German
- **Live app:** <https://project-brief-nlp-automated-customers-reviews-4z3nd6dqfyj7ba8x.streamlit.app/>
- **Code and README:**
<https://github.com/alkilicayse87-pixel/project-brief-NLP-automated-customers-reviews>

Prezi: insert the 2-3 min app video here. Narration script is on the next slide.

App Video Narration (about 2:30) - preparation only, skip when p

0:00 This is our customer review intelligence app, built on 34,625 Amazon reviews and deployed publicly on Streamlit Community Cloud.

0:20 Dashboard. Pick a product category. We see the sentiment mix, review length, the products with the most negative reviews and the main complaint themes. We can switch between our model's predictions and the star-rating labels.

1:00 Predictor. Paste any review and the model returns negative, neutral or positive with scores. Because the data is imbalanced, it is strongest on positive reviews, which is why we report macro F1.

1:30 Translate to German. Paste an English review, for example 'Great tablet, works without wifi, love it!'. We get a German translation, product names stay unchanged, and sentiment is preserved in 80% of our test cases.

2:20 No review text is published, only aggregated data. The live link and the code are on the screen. Thank you.

Business Value: From Reviews to Decisions

- **Before:** thousands of reviews, manual analysis, time consuming
- **After:** our NLP pipeline automatically detects sentiment, groups similar products, generates AI summaries and recommendations, and translates reviews into German
- **Outcome:** customer feedback becomes actionable and easier to interpret
- **Not just models:** everything runs as a public Streamlit app anyone can try

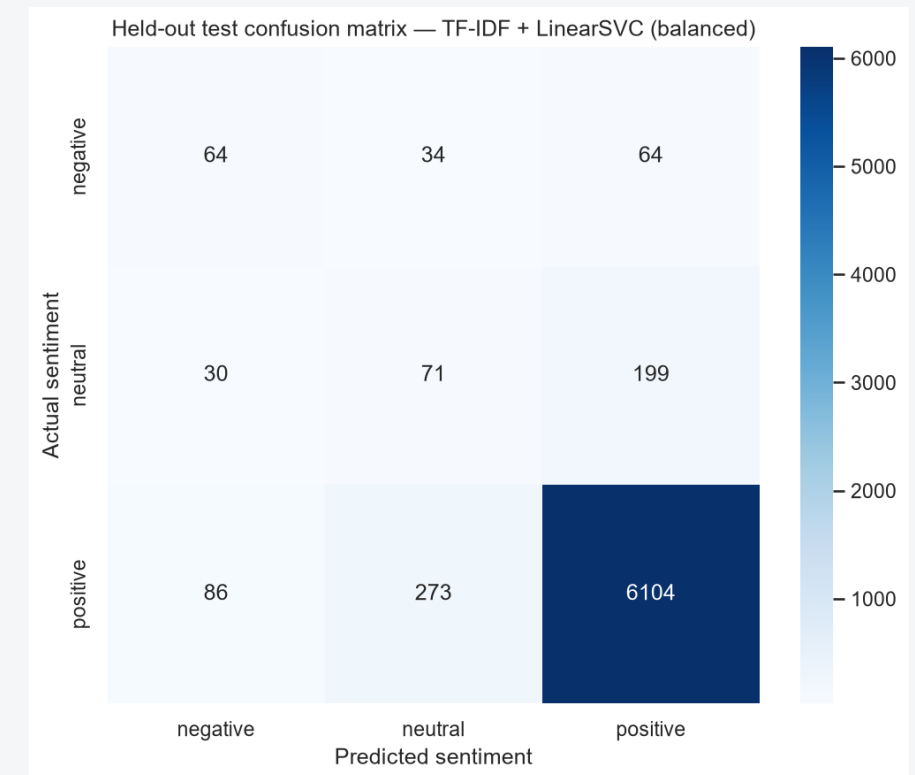
Teamwork and Project Management

- **How we worked:** together through Tasks 1-5; reviewed each milestone before moving on; tested alternative solutions independently; chose the final implementation together
- **Lessons learned:** continuous communication improved decisions; independent experiments added valuable improvements; better planning for optional features would have saved time
- **Risk management:** no raw review text or secrets published; heavy models load on demand; results judged on macro F1, not accuracy

Say (about 20 s): "Although we had different areas of focus, we worked through all project stages together. We reviewed each milestone before moving forward, tested alternative approaches independently, compared our results, and jointly selected the final solution for deployment and presentation."

Major Obstacle

- **Initial assumption:** 90% accuracy means excellent performance
- **Reality:** 92% of all reviews are positive
- **Result:** the model predicts positive reviews very well but struggles with neutral (F1 21%) and negative (F1 37%) ones
- **Lesson:** always evaluate imbalanced data with class-based metrics such as macro F1



In Hindsight

- Write the hypothesis and success metrics on day one
- Fine-tune a transformer and calibrate scores instead of TF-IDF + SVM only
- Keep a raw data layer; do not overwrite text with the cleaned version
- Check summaries against the source reviews automatically
- Have a native German speaker rate about 100 translations

Key Takeaways: Was the Hypothesis Supported?

- **What worked:** sentiment accuracy 90% (positive F1 95%); 5 meaningful product clusters; AI summaries gave useful buyer guidance; German translation kept sentiment in 80% of cases
- **Where it fell short:** negative (F1 37%) and neutral (21%) detection; cluster silhouette only ~0.11
- **What we learned:** accuracy misleads on imbalanced data; humans must validate GenAI output; NLP can ship as a practical business tool
- **Hypothesis:** supported for positive sentiment, summaries and translation; only partly for clustering and negative/neutral sentiment

Insights, Implications and Open Questions

- **Surprise:** a high headline accuracy can come almost entirely from class imbalance
- **Implication:** product teams can scan complaint themes by category quickly, but should verify before acting
- **Open questions:** how noisy are star-rating labels? Does sentiment change over time (the data has no dates)? How good is the German according to native speakers?

Thank You

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